Complete papers and shared theories

Thirteen topics, thirty-four original texts

This candidate overview combines eleven unchanged admitted topic rows with the new unified B edition and the enlarged N007–N006 orbit-construction edition. Both new numerical packets have distinct frozen root acceptance; this consolidated overview still awaits its separate review. Their exact result hashes are bound directly; no future public commit is invented. The unchanged baseline is committed `ea98753`; both earlier overviews remain preserved.

For each topic, $S = \sum_i c(L_i)$ counts its complete original texts, U every complete common paper, and $A = \sum_i c(A_i)$ every complete application. The comparisons are

$$R = \frac{S}{U + A}, \qquad r_i = \frac{c(L_i)}{c(A_i)} \quad \text{after acquiring the whole theory and declared software/data.}$$

Each topic pays its common theory independently, even when proofs overlap another topic. A factor above one means a shorter replacement in that coordinate; a factor below one means a longer one. Conditional factors do not include acquisition a second time, and are not standalone application savings.

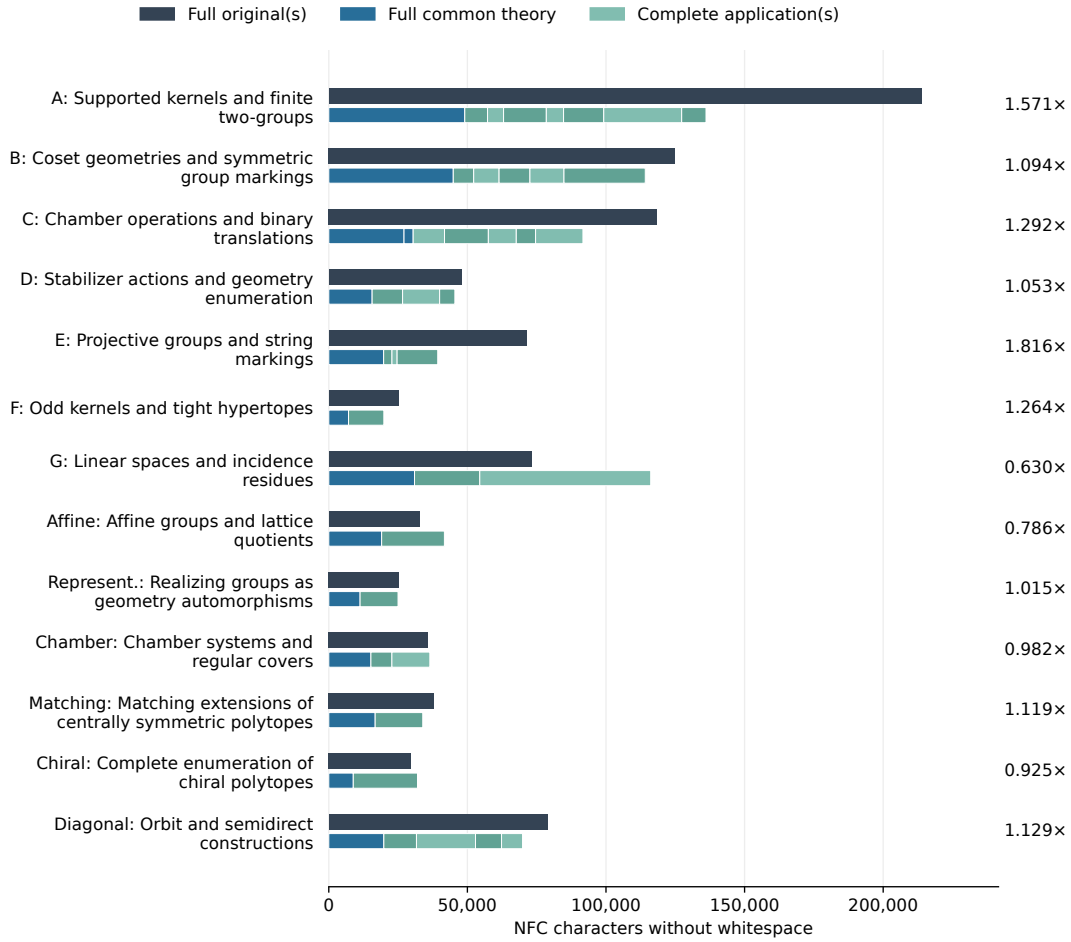
Complete written costs. All rows use NFC characters without whitespace at their stated representations. Full-document boundaries include introductions, context, proofs and references.

Topic	Originals	Sources	Common	Applications	Written	Factor
A	7	213,803	49,059	87,019	136,078	1.571
B	5	124,906	44,905	69,265	114,170	1.094
C	5	118,433	30,437	61,248	91,685	1.292
D	3	47,894	15,632	29,855	45,487	1.053
E	3	71,375	19,790	19,513	39,303	1.816
F	1	25,115	7,118	12,744	19,862	1.264
G	2	73,199	30,928	85,207	116,135	0.630
Affine	1	32,767	19,070	22,638	41,708	0.786
Represent.	1	25,348	11,288	13,688	24,976	1.015
Chamber	2	35,802	15,199	21,254	36,453	0.982
Matching	1	37,951	16,732	17,168	33,900	1.119
Chiral	1	29,599	8,831	23,171	32,002	0.925
Diagonal	2	78,889	19,859	50,036	69,895	1.129

Representations remain distinct. B and C use complete authored-text renderings for N001 and L108, with raw untrimmed PDFs for their other documents. Those originals' unavailable-PDF costs remain unknown. Other rows use their accepted selected, calibrated PDF coordinate. Diagonal's encoded-event graphical proxies retain raw and text-only sensitivities. No pooled factor is formed.

Scope. This is a proposed numerical account of 34 original texts within the 161-item programme, not completion of that programme. The committed scientific snapshot has 34 recoveries among 104 accessible originals, with 70 awaiting complete recovery and 57 retrieval targets. All corrections, retained premises and ancillary exclusions remain. No scientific, source, question or formal credit changes.

Complete common theories and applications. Dark bars are whole original sums. Blue segments are whole common papers; green segments are all complete applications. Unified B pays its entire current common paper once. C still pays its complete additional incidence note. The N006 application is displayed as all three companion-paper segments.



The enlarged Diagonal row now contains both N007 and N006 and the whole current eight-page theory. It uses the new frozen two-original measurement; no factor or common cost is borrowed from the old N007-only edition. Text length is not a minimum proof complexity or a measure of originality.

Complete programs and prior acquisition. The written comparison remains first. These separate alternatives add the entire declared program representation to the written denominator. Raw and printed forms are never added together.

Topic	Printed code	With printed	Raw software	With raw	U+programs
A	54,210	1.124	52,807	1.132	103,269
B	13,170	0.981	12,330	0.987	58,075
C	20,933	1.052	19,937	1.061	51,370
D	27,070	0.660	26,185	0.668	42,702
E	7,565	1.523	–	–	27,355
F	8,837	0.875	8,061	0.899	15,955
G	53,018	0.433	50,184	0.440	83,946
Affine	17,250	0.556	16,682	0.561	36,320
Represent.	9,025	0.746	8,310	0.762	20,313
Chamber	7,677	0.811	6,401	0.835	22,876
Matching	4,998	0.976	4,408	0.991	21,730
Chiral	11,940	0.674	11,090	0.687	20,771
Diagonal	–	–	394,294	0.170	414,153

The final column uses the complete printed form when available. For Diagonal it uses all nineteen raw program/witness files plus both required decoding drivers. Its complete printed route is unavailable: the old two-page appendix covers only one validation file. Diagonal conditionals also assume acquisition of all required external data, separately inventoried as 21,960,689 representation bytes and 2,516,249 listed installed bytes. Byte storage is not added to character units. Large witnesses are not discounted.

E retains its separate optional-validation sensitivity: another 5,173 characters gives a program-inclusive factor of 1.372. No unadmitted raw-code alternative is invented for E.

Roles. A includes five full files. B includes a finite premise, two validations and an auxiliary example. C includes its three complete GAP files, counting the identical shared construction once within C and retaining its exhaustive census. D includes five required and three diagnostic programs. E separates its required program from optional validation. F's three programs and Representation's two are validations. G retains four finite computations, two certificate bridges and seven validations. Affine retains six mathematical programs; Chamber one required and one diagnostic program; Matching its census; Chiral enumeration all three files. Diagonal includes sixteen finite proof/witness files, three validations and both required decoders. Other operational drivers and partial Lean inventories stay separate.

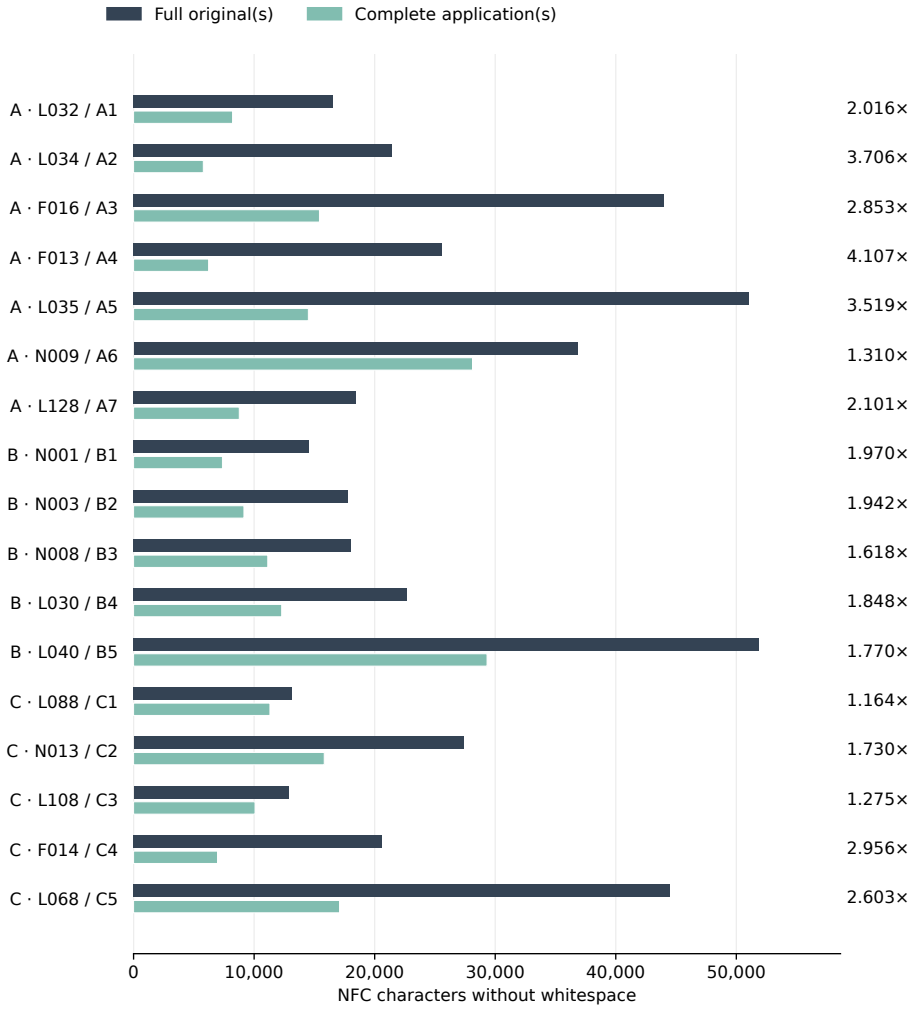
Limits. Complete external proofs, unprovided historical software, full graphics and end-to-end formalization have unknown costs, not zero. Every topic retains its underlying multi-unit and source-cover sensitivities. B's source costs are unchanged from the prior edition. This table does not compare a theorem closure with a complete original or use a node-count ratio.

Every complete application after full acquisition. All common papers and all declared software/data have already been acquired. The whole application is charged, including local context and references. DG2 includes every N006 companion paper.

Topic	Original	Application	Source cost	Application cost	Conditional
A	L032	A1	16,558	8,212	2.016
A	L034	A2	21,433	5,784	3.706
A	F016	A3	43,983	15,419	2.853
A	F013	A4	25,562	6,224	4.107
A	L035	A5	51,019	14,498	3.519
A	N009	A6	36,824	28,114	1.310
A	L128	A7	18,424	8,768	2.101
B	N001	B1	14,519	7,371	1.970
B	N003	B2	17,780	9,155	1.942
B	N008	B3	18,009	11,132	1.618
B	L030	B4	22,704	12,286	1.848
B	L040	B5	51,894	29,321	1.770
C	L088	C1	13,166	11,311	1.164
C	N013	C2	27,361	15,815	1.730
C	L108	C3	12,849	10,075	1.275
C	F014	C4	20,577	6,962	2.956
C	L068	C5	44,480	17,085	2.603
D	L066	D1	20,916	10,989	1.903
D	L074	D2	20,661	13,367	1.546
D	L104	D3	6,317	5,499	1.149
E	L083	E1	20,885	3,002	6.957
E	L079	E2	20,908	1,855	11.271
E	L021	E3	29,582	14,656	2.018
F	L014	F1	25,115	12,744	1.971
G	L009	G1	49,891	23,544	2.119
G	L092	G2	23,308	61,663	0.378
Affine	L012	AF1	32,767	22,638	1.447
Represent.	F010	R1	25,348	13,688	1.852
Chamber	L008	H1	21,784	7,571	2.877
Chamber	N016	H2	14,018	13,683	1.024
Matching	F015	M1	37,951	17,168	2.211
Chiral	L028	CE1	29,599	23,171	1.277
Diagonal	N007	DG1	22,888	11,771	1.944
Diagonal	N006	DG2	56,001	38,265	1.464

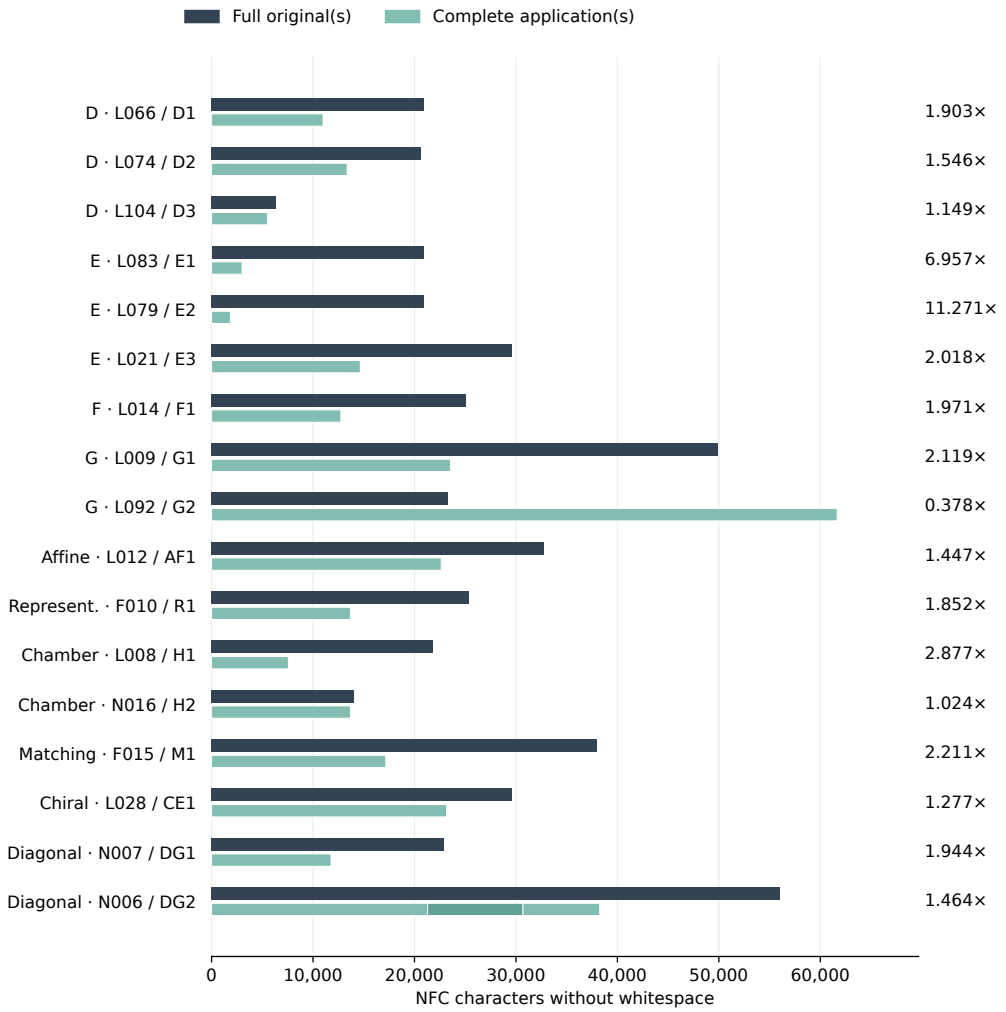
The B and Diagonal rows use their exact newly accepted numerical packets. Other rows remain unchanged from the committed admitted overview. This consolidation adds no scientific or numerical admission.

Individual source and application costs: first seventeen originals. Dark bars show complete original texts; green bars show complete applications. Factors at right assume full prior acquisition. The source and replacement bars share the same scale within each chart.



Source IDs are stable catalogue cross-references. Descriptive topic titles, exact editions and current source-contract qualifications remain in the bound records; they are not new claims of this overview.

Individual source and application costs: remaining seventeen originals. The same conventions apply. The final N006 bar includes its general application, nine-case certificate paper and Harada–Norton companion, with all references retained.



Data and provenance. Exact integers and rational comparisons are in `results.json`, `topics.csv` and `applications.csv`. The protocol pins committed unchanged numerical/admission records and the two new numerical packets and their distinct acceptance records directly. No primary text is copied or extracted. Bar segments use complete document costs, not theorem counts. Distinct review is required before this overview becomes the public current edition.